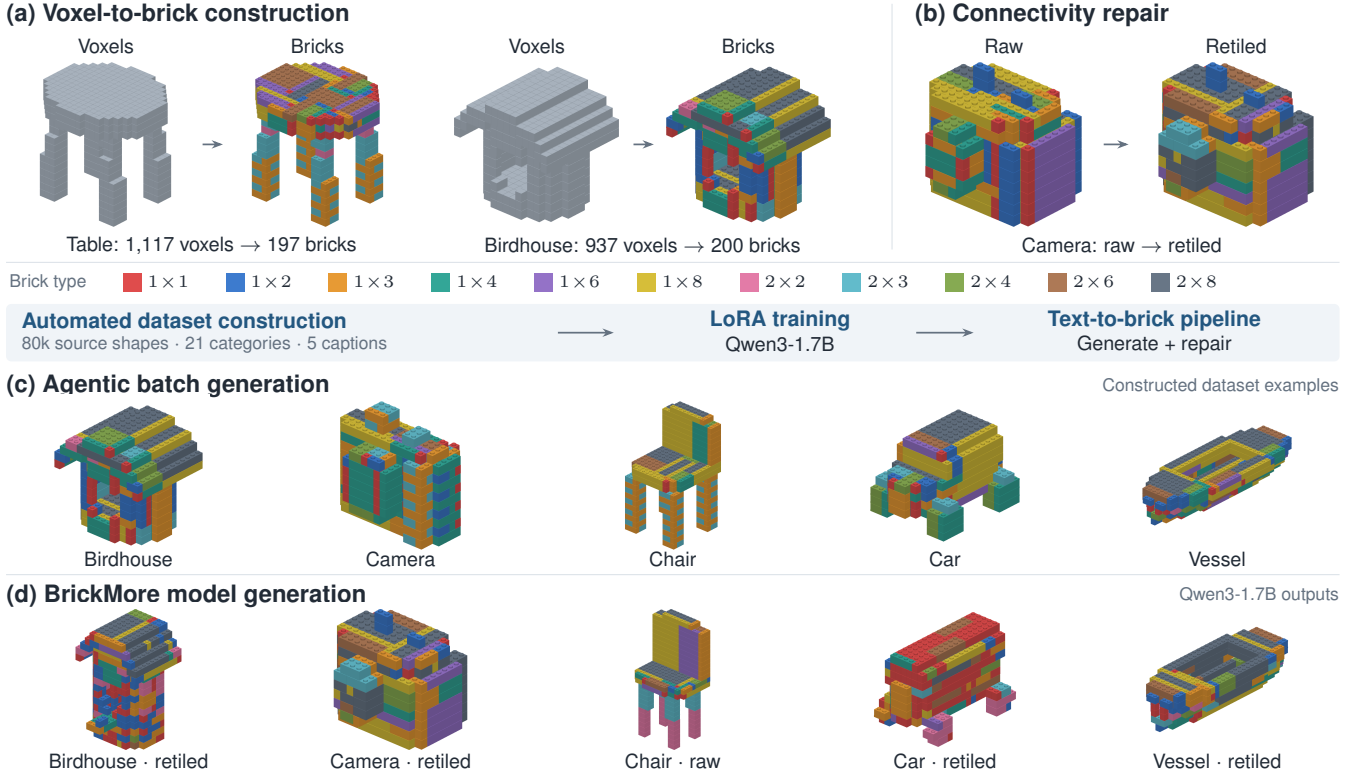

(a) Voxel-to-brick construction
Voxels
Bricks
Voxels
Bricks
Table: 1,117 voxels → 197 bricks
Birdhouse: 937 voxels → 200 bricks
(b) Connectivity repair
Raw
Retiled
Camera: raw → retiled
Brick type
1×1
1×2
1×3
1×4
1×6
1×8
2×2
2×3
2×4
2×6
2×8
Automated dataset construction
80k source shapes · 21 categories · 5 captions
LoRA training
Qwen3-1.7B
Text-to-brick pipeline
Generate + repair
(c) Agentic batch generation
Constructed dataset examples
Birdhouse
Camera
Chair
Car
Vessel
(d) BrickMore model generation
Qwen3-1.7B outputs
Birdhouse · retiled
Camera · retiled
Chair · raw
Car · retiled
Vessel · retiled